

The Duke Lung Cancer Screening (DLCS) Dataset: A Reference Dataset of Annotated Low-Dose Screening Thoracic CT

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See also commentary by Hsu and Prosper in this issue.

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Lung cancer is the leading cause of cancer-related death worldwide (1). Randomized clinical trials demonstrated that low-dose CT (LDCT) screening reduces the risk of mortality (2,3). Development of automated systems may facilitate LDCT screening, but curated imaging data are limited due to the time-consuming nature of manual annotations. Existing datasets for lung nodule classification (4–7), such as the Lung Image Database Consortium and Image Database Resource Initiative, are over a decade old and no longer reflect current CT scanner technology or clinical practice (8); others lack bounding box lesion annotations, such as the National Lung Screening Trial (2).

This study presents a dataset of screening CT scans from a high-volume contemporary lung cancer screening center with semiautomated annotations for clinically relevant lung nodules. In addition, the dataset includes Lung CT Screening Reporting and Data System (Lung-RADS) scores and, if applicable, timing and histologic type and stage of future lung cancer. Such a dataset could be valuable for lung cancer risk stratification, radiomics analysis, and development of artificial intelligence (AI) algorithms (9–11).

Materials and Methods

Patient Sample

This Health Insurance Portability and Accountability Act–compliant retrospective study was approved by our institutional review board (Pro00108863) with waiver of informed consent. We identified patients receiving LDCT screening from January 1, 2015, through June 30, 2021. Lung-RADS scores for each LDCT examination were extracted from the radiologist report; Lung-RADS versions 1.0 and 1.1 were in clinical use during this study period. Changes in version 1.1 included increasing the size threshold of nonsolid nodules and reclassification of perifissural nodules less than 10 mm to category 2 (12). If unavailable, a Lung-RADS score was assigned by a fellowship-trained cardiothoracic radiologist using version 1.1 criteria. Patient characteristics were extracted from our electronic health record and our center's tumor registry. A prior study examined predictors of lung cancer diagnosis for screening patients during the study period and provided characteristics of the broader screening population at our institution (13). In the current study, we focus on the subset with complete image and report data who met our nodule annotation criteria. For patients with multiple LDCT scans in the study period, we used the terminal examination. Scans were excluded if they were Lung-RADS 0, image

or report data were unavailable, or no nodules met the annotation criteria (described below).

Annotation of Lung Nodules

The annotation process was performed semiautomatically by a medical student (A.J.W.) and overseen by fellowship-trained cardiothoracic radiologists with 9 (T.D.T.) and 2 (M.R.H.) years of experience. Annotations were performed for all potentially actionable nodules mentioned in the radiologist report with a size of 4 mm or greater or any segmental or proximal airway nodule, as these could potentially constitute a positive screening examination (Lung-RADS 3, 4A, 4B, or 4X).

The MONAI nodule detection algorithm was previously evaluated on the LUNA16 dataset, yielding cross-validation sensitivities of 0.835 and 0.988 at one of eight and eight false positives per scan, respectively (14). For this study, the algorithm provided initial nodule candidates (pseudo-labels) as a three-dimensional bounding box and detection score ranging from 0 to 1 (14). A custom Python program selected the highest scoring nodule candidate matching the reported section number if the score was 0.1 or greater and the bounding box size was within 50% of the reported size. If all nodules in a scan were successfully annotated, the scan was placed in annotation group 1. Scans without nodule candidates with a score of 0.1 or greater on a section of interest but containing other annotations consistent with the reported Lung-RADS score were placed in group 2.

If the most concerning nodule in a positive screen was not successfully labeled with the above process, it was manually labeled by the medical student alone, who confirmed the low-scoring nodule annotation by the algorithm (group 3), or manually annotated by a radiologist or by the medical student with subsequent radiologist confirmation and/or correction (group 4). We excluded negative screening examinations with nodules that met our criteria but no appropriate preliminary annotations. All semiautomated annotations were inspected to ensure a pulmonary lesion was annotated; obvious failures were removed.

Dataset Construction

Axial LDCT images with section thickness 0.6 mm for included scans were de-identified to remove protected health information and stored as Neuroimaging Informatics Technology Initiative files. Further imaging protocol information can be found in a prior study (13). The annotations are object (“.obj”) files, which contain eight coordinates representing a three-dimensional

Abbreviations

AI = artificial intelligence, LDCT = low-dose CT, Lung-RADS = Lung CT Screening Reporting and Data System

Summary

The Duke Lung Cancer Screening (DLCS) dataset is a large collection of lung cancer screening low-dose CT scans for lung nodule classification with annotations performed in a semiautomated manner, requiring substantially reduced radiologist effort.

Key Points

- The Duke Lung Cancer Screening set contains 2061 lung cancer screening low-dose CT scans, acquired within the past 10 years, with 3187 semiautomatically annotated nodules.
- This image-rich dataset also includes Lung Imaging Reporting and Data System scores, thus serving as a valuable resource for future research aimed at refining pulmonary nodule classification.
- The reproducible method used for dataset creation is efficient and may be used to annotate similar datasets in the future.

Keywords

CT, Thorax, Lung, Neoplasms-Primary, Oncology

bounding box for each nodule. The following information was recorded in a spreadsheet: how each scan was annotated, patient demographic information, Lung-RADS score, Lung-RADS version used, minimum 1-year follow-up (yes or no), and biopsy-proven lung cancer diagnosis timing, if applicable.

Results

Patient Characteristics

The dataset included 2061 LDCT scans containing 3187 annotated nodules (Figs 1, S1). The included patients (1032 female, 1029 male) had a mean age at time of screening of 66.7 years $\pm$ 6.2 (SD). Demographic characteristics are reported in Table 1, and clinical characteristics are reported in Table 2.

LDCT Annotations

Figure 2 describes the annotation process and illustrates how LDCT images were annotated. The majority of scans were annotated semiautomatically (85.8% of scans [1768 of 2061]; 84.1% of nodules [2681 of 3187]). Among all semiautomated annotations, only 22 of 2222 (1.0%) of nodules in group 1 and 11 of 477 (2.3%) of nodules in group 2 failed to capture a pulmonary lesion. These 33 failures were removed from the dataset.

Each subset was spot-checked by a cardiothoracic fellowship-trained radiologist using a simple random sample. Success required that the majority of the nodule from the radiologist report, including its center of volume, be contained within the annotation. Successful results were confirmed for 33 of 34 annotations (97%) across 25 scans from the fully automatically labeled group 1, 32 of 33 annotations (97%) across 25 scans from group 2, and 45 of 45 annotations (100%) across 25 scans for group 3. All group 4 annotations that were initially performed by the medical student were reviewed by a radiologist, who confirmed accuracy for 180 of 183 (98%) annotations across 110 scans and corrected the remainder. Examples of annotations are shown in Figure 3.

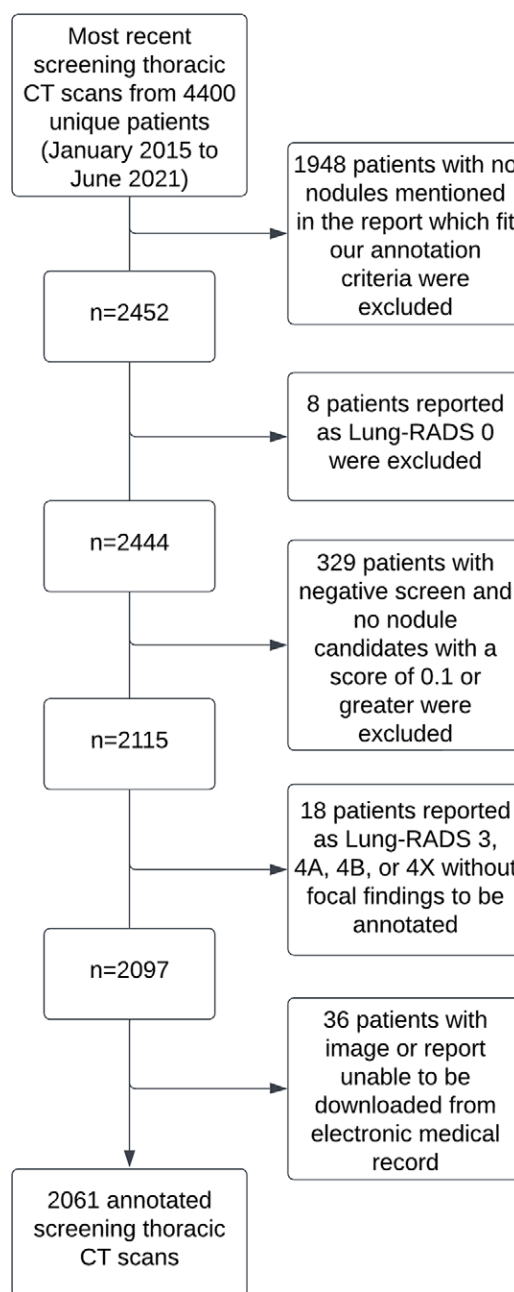


Figure 1: Diagram describes selection of low-dose CT examinations. Lung-RADS = Lung CT Screening Reporting and Data System.

Data Availability

Full CT images, annotations, and basic demographic and clinical characteristics for 1613 LDCT scans with 2487 annotated nodules are publicly available at <https://doi.org/10.5281/zenodo.10782890>. The other 448 scans (21.7%, 448 of 2061) are withheld as test cases, such as for a data challenge.

Discussion

The development of large collections of CT scans with annotations of lung nodules is an important step in furthering medical imaging research. The DLCS dataset serves to bridge this gap with images acquired within the past 10 years. Including Lung-RADS scores and future lung cancer diagnoses in our dataset

Table 1: Patient Demographics

Demographic	No. of Patients (<i>n</i> = 2061)
Age at time of screening (y)*	66.7 (6.2)
Age at time of screening (y) [†]	67 (62–71)
Sex	
Female	1032 (50.1)
Male	1029 (49.9)
Race	
Asian	18 (0.9)
Black or African American	476 (23.1)
White	1519 (73.7)
Not reported/declined	24 (1.2)
Other [‡]	24 (1.2)
Ethnicity	
Hispanic or Latino	11 (0.5)
Not Hispanic or Latino	1981 (96.1)
Not reported	69 (3.3)
Marital status	
Divorced/legally separated	250 (12.1)
Married	1053 (51.1)
Single/life partner	370 (18.0)
Unknown	37 (1.8)
Widowed	351 (17.0)
Insurance status	
Commercial	752 (36.5)
Medicaid	74 (3.6)
Medicare	1192 (57.8)
Other government (VA or Tricare)	14 (0.7)
Self	29 (1.4)
Smoking status	
Current	1043 (50.6)
Former	903 (43.8)
Unavailable	115 (5.6)

Note.—Data are expressed as numbers of patients, with percentages in parentheses, unless otherwise indicated. VA = Veterans Affairs.

* Value is the mean, with SD in parentheses.

[†] Value is the median, with IQR in parentheses.

[‡] Includes patients whose self-reported race was “American Indian or Alaskan Native,” “Native Hawaiian or Other Pacific Islander,” “two or more races,” or “Other.”

provides additional clinical information about lung cancer risk and subsequent management decisions. The most readily apparent application of this dataset is testing and training of AI algorithms to classify malignancy.

Our semiautomated annotation method has the potential to be used to efficiently generate similar datasets in the future. Aided by weak labels in the radiology report text, the medical student annotations were confirmed by radiologists to be highly accurate, which could be an avenue to further reduce radiologist annotation effort. The development of similar datasets across institutions would enable data pooling and improved performance of models.

The lung nodule annotations in our dataset are not intended to be comprehensive. Radiologist report text was used to determine clinical relevance, and it is expected that many tiny or clearly benign nodules would not be individually reported. The exclusion of

Table 2: Patient Clinical Characteristics

Clinical Characteristic	No. of Patients (<i>n</i> = 2061)
Total scans	2061 (100)
Lung-RADS category	
1	11 (0.5)
2	1145 (55.5)
3	363 (17.6)
4A	286 (13.9)
4B	188 (9.1)
4X	68 (3.3)
Lung cancer diagnosis time from LDCT	
Within 1 year	118 (5.7)
Between 1 and 2 years	18 (0.9)
2 years or greater	19 (0.9)
Total	155 (7.5)
Lung cancer type (<i>n</i> = 155)	
Adenocarcinoma	67 (43.2)
Large cell carcinoma	1 (0.6)
Non-small cell lung cancer	19 (12.3)
Other	4 (2.6)
Small cell carcinoma	17 (11.0)
Squamous cell carcinoma	33 (21.3)
Unknown	14 (9.0)
Cancer stage (<i>n</i> = 155)	
Stage I	69 (44.5)
Stage II	19 (12.3)
Stage III	19 (12.3)
Stage IV	34 (21.9)
Unknown	14 (9.0)

Note.—Data are expressed as numbers of patients, with percentages in parentheses. LDCT = low-dose CT, Lung-RADS = Lung CT Screening Reporting and Data System.

many scans that did not meet annotation criteria likely explains the relatively high percentage, at 7.5% (155 of 2061), of patients in our dataset who had later diagnoses of lung cancer. We used the terminal examination to ensure the study preceding a cancer diagnosis, if applicable, was included in our dataset.

Several limitations should be acknowledged. First, the algorithm used for generating nodule candidates occasionally over- or underestimated the size of the nodule. Second, Lung-RADS has undergone three iterations over the study period, and the nodules included in this dataset were categorized using Lung-RADS versions 1.0 and 1.1, which were used in clinical practice at the time these scans were acquired. Post hoc reinterpretation is not practicable and may not apply to future changes in standard reporting guidelines (15). Third, variability between radiologists performing the clinical interpretation may affect which nodules were reported and thus included (16). Finally, as with all data curated in a semiautomated manner, there will be some degree of error.

In summary, the DLCS dataset represents a useful resource that reflects the technologies of a contemporary LCS program, and the methods used for its design may allow improved efficiency and scale of future dataset creation.

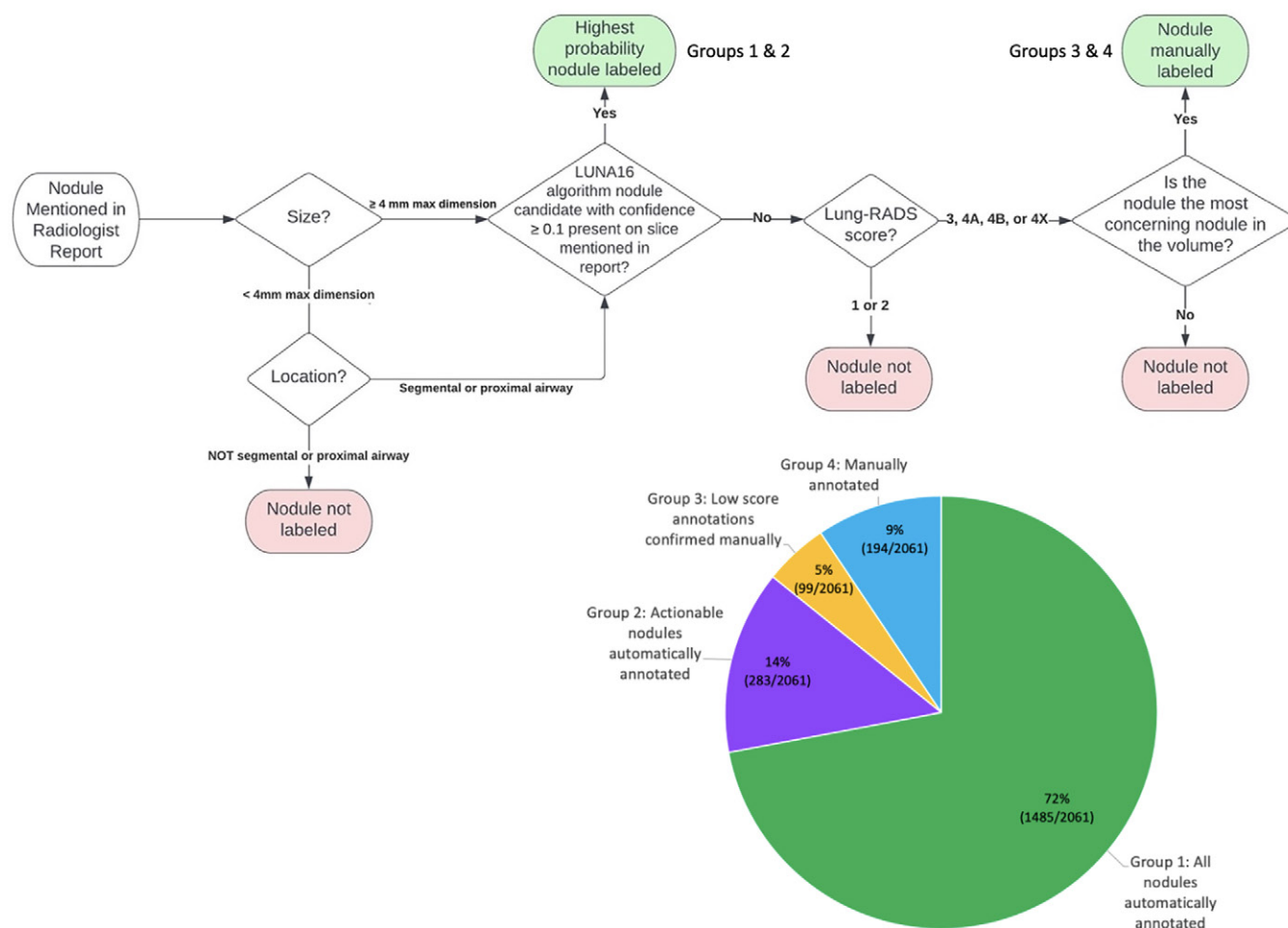


Figure 2: Flowchart demonstrates the annotation decision-making process for each nodule and pie chart illustrates annotation methods used to generate the dataset, by scan. Lung-RADS = Lung CT Screening Reporting and Data System.

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Data sharing: Data generated by the authors or analyzed during the study are available at: <https://doi.org/10.5281/zenodo.10782890>.

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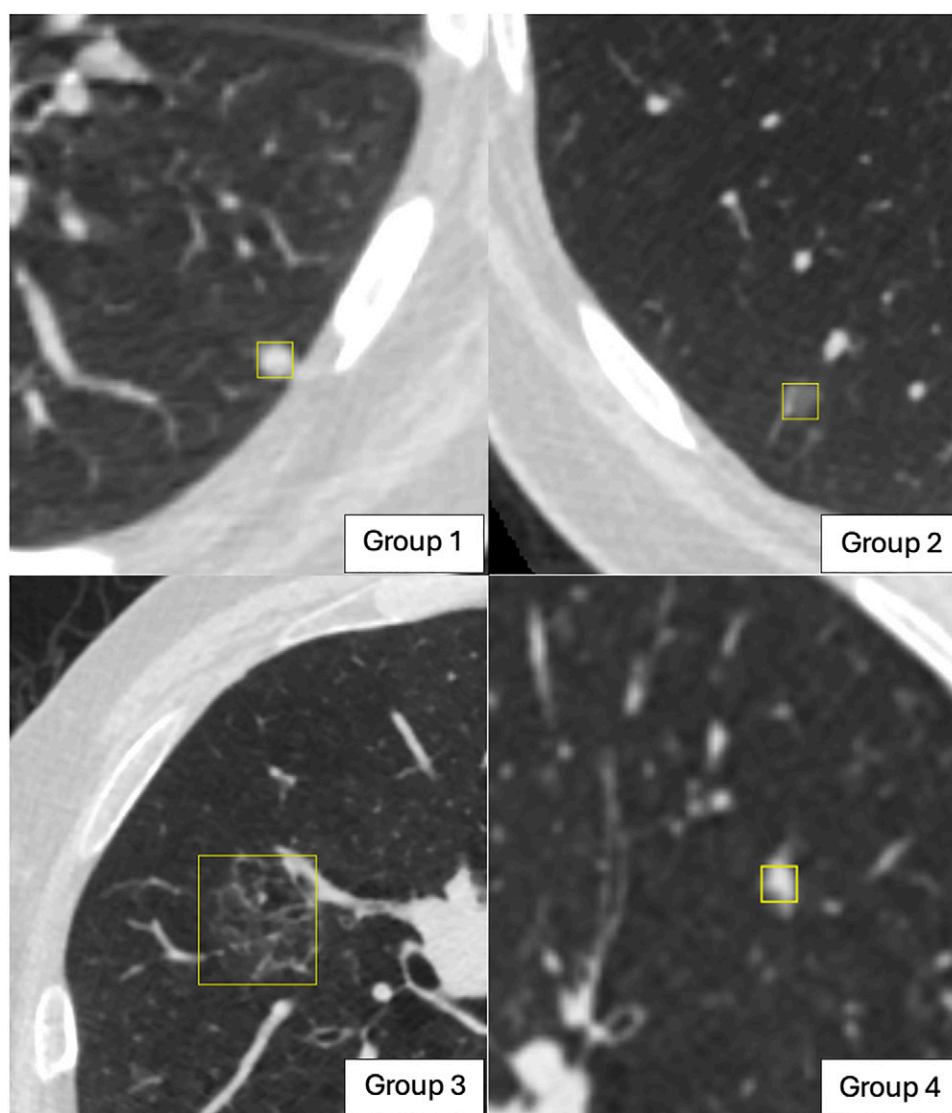


Figure 3: Examples of lung nodule annotations in the Duke Lung Cancer Screening dataset, representing groups 1–4. Annotations in groups 1 and 2 were annotated entirely by the automated algorithm, annotations in group 3 were annotated by the automated algorithm and manually confirmed, and annotations in group 4 were performed manually. Annotations are in the form of three-dimensional bounding boxes (yellow squares), though only a single section at axial CT is shown here. An example excerpt of the report text used to identify the group 4 nodule above is as follows: "...There are scattered pulmonary nodules, most of which likely represent mucous plugs, the largest of which measures 11.9 × 6.6 mm (series 4, image 222) with a mean diameter of 9.2 mm... ACR Lung-RADS Category and Recommendation*: 4AS." ACR = American College of Radiology.

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